Comparative Corpus Analysis: Temporal Evolution of AI/ML/NLP Research

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Abstract

In this research we investigate the nature of drift in AI/ML/NLP research over a 50-year period (1990-2021). We built a text processing pipeline with two key experimental factors: (1) text features (TF-IDF unigrams, bigrams, and SBERT embeddings) and (2) comparison strategies (top terms overlap, cosine similarity, and temporal classification). Analysing 130,279 abstracts, we observe semantic embeddings (SBERT) better represent key concepts than text features, while bigram representations are most responsive to a change in vocabulary and approach. We found an important break in 2007 which we consider to be the shift from symbolic to statistical/deep learning. In the last few years (2013-2021), we observe convergence of vocabulary to key concepts of deep learning. Our research shows the need for multiple strategies to analyse text over time, as they provide different perspectives on research evolution.

Introduction

Changes in research topics and approaches can provide insights into research advances. The fields of artificial intelligence, machine learning and natural language processing have evolved rapidly in recent decades from symbolic rule-based to statistical learning approaches, and now more recently, massive deep learning models.

In our coursework we explore the research question:

How has research in AI, ML and NLP in terms of topics, communication and discourse evolved from 1990 to 2021 and how do different text analytics techniques provide different perspectives?

There are problems with comparing this evolution:

- • Class imbalance: there are many more papers in recent decades (23,092 in 2018-21) than in the first decades (115 in 1990-99), making quantitative comparisons difficult.
- • Concept evolution: terms change - 'neural networks' in 1990s and 'transformers' and 'attention mechanisms' in 2020s are different computer models, and mean different ways of implementing the same idea.
- • Methodology differences: different ways to represent and compare text show different aspects of drift, and are all valid.

To address these problems we employed a scientific approach, by hypothesising numerous different text representations and different ways of comparing text, and testing these to learn how each has their own merits. We have 130,279 abstracts from arXiv for 1990-2021, in categories cs.AI, cs.LG, cs.CL, cs.NE, and stat.ML.

We divided this corpus into five periods (1990–99, 2000–06, 2007–12, 2013–17, 2018–21) based on key inflection points in the field's history.

Methods

Data and Preprocessing

We have a corpus of 130,279 arXiv papers abstracts in five categories:

cs.AI, cs.LG, cs.CL, cs.NE, and stat.ML.

We applied standard NLP preprocessing to normalize words: we converted all words to lower case, removed punctuation and English stopwords (NLTK English stopword list) and lemmatized words

(WordNetLemmatizer). This ensures the data is prepped, and its meaning is preserved. However, we also note that lemmatization removes morphological differences, which could mask vocabulary differences that indicate method change.

We divided the data into five periods of interest:

- • 1990-99: Early years with few papers (n=115)
- • 2000-06: Statistics are recognised (n=331)
- • 2007–12: Emergence of deep learning (n=1,489)
- • 2013–17: Deep learning dominance (n=6,050)
- • 2018–21: Modern era with transformer-based models (n=23,092)

Axis 1: Text Representation

We anticipated that different text representations would reveal different patterns of change in research themes. We compared three representations: TF-IDF Unigrams We applied TF-IDF (term frequency-inverse document frequency) to words using scikit-learn's TfidfVectorizer (with a maximum of 15,000 features). TF-IDF considers the significance of a term to the document and the rarity of the term within the corpus, meaning that common domain-specific terms are favoured. This emphasises vocabulary shift.

TF-IDF Bigrams

We used TF-IDF on these word pairs such as "deep learning", "attention mechanism" and "neural network". Bigrams are more specific research techniques and concepts than unigrams. We hypothesised that the bigram drift would be more than the unigram drift because research language evolves at the term level, rather than the word level.

SBERT Embeddings

We transformed each of the abstracts into a 384-dimensional semantic embedding using a pre-trained transformer called Sentence-BERT (SBERT, all-MiniLM-L6-v2). SBERT produces a good measure of similarity between sentences, capturing the ideas and not words. We hypothesised that semantic embeddings will be less likely to drift than TF-IDF variants because papers discussing the same topic may use different words. We used the average cosine distance within each period as a "drift measure". This measures the change in research topics.

Axis 2: Comparison Methods

We compared three methods of representing change, using the same approach for representing the text (TF-IDF bigrams which were most drifty):

Method 1: Top-Term Overlap

We took the top 10 most important terms (as per the TF-IDF weight) in each period. We then determined what percentage of the top 10 terms for period t were also in the top 10 for period $t+1$. This indicator is a measure of the consistency of vocabulary and the ideas being discussed.

Method 2: Cosine Similarity Matrix

We first computed the average cosine similarity between all pairs of abstracts for each period, and then the cosine similarities between periods. This gives us a similarity matrix for all periods that is symmetric. This measure of similarity is between 0 (no overlap) and 1 (overlap exactly).

Method 3: Classify periods with Linear SVM

We classified abstracts into periods with a linear support vector machine (SVM). We trained this SVM on 80% of the data, with class weights adjusted for the class imbalance. We used the remaining 20% of the data as a test set, and reported precision, recall, F1-score and confusion matrices. If the periods are different in vocabulary, the accuracy of classification will be high; if the drift is smooth, the accuracy of classification will be low.

Evaluation Metrics We evaluated the methods in three ways:

- • Soundness: Does the result reflect the key changes in the field?
- • Interpretability: Do we get any insights into the change?
- • Known patterns: Are results consistent with history of the fields?

GitHub repository: https://github.com/AmritRajeev/Text_AI_Assignment

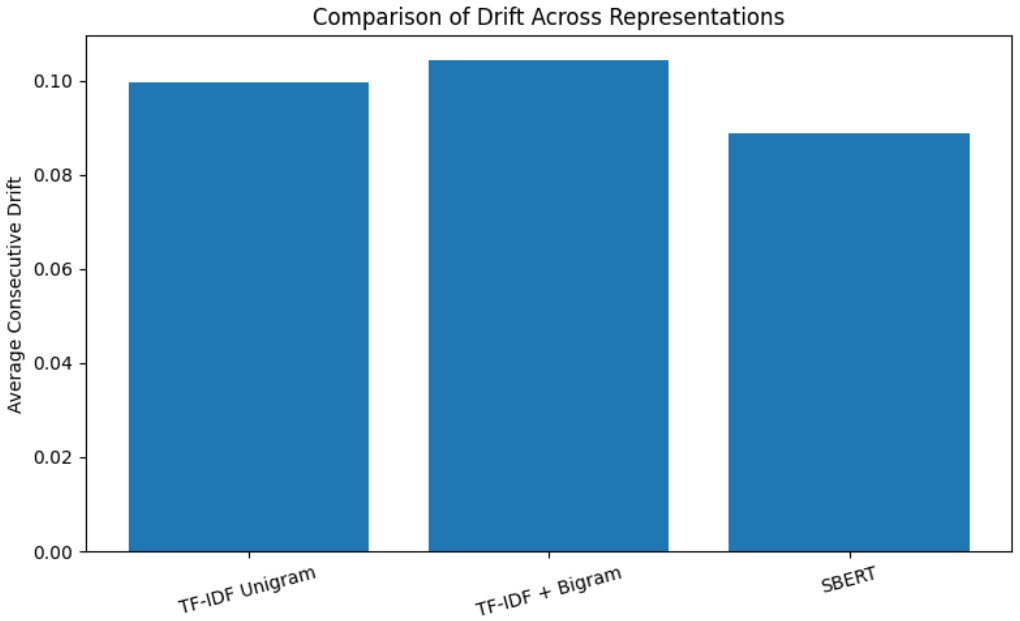
Experimental Evaluation

Results: Axis 1 (Text Representation)

We computed the average cosine distance between consecutive periods for each representation:

Representation	**Avg Consecutive Drift**
TF-IDF unigram	0.0995
TF-IDF unigram+bigram	0.1043
SBERT embeddings	0.0888

TABLE 1: Comparison of Drift Across Text Representations. TF-IDF bigrams show the highest drift (0.1043), indicating that phrase-level terminology changes fastest. SBERT embeddings show the lowest drift (0.0888), demonstrating that underlying semantic concepts persist across periods despite surface vocabulary changes.



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Both hypotheses were confirmed:

Bigrams showed the highest drift (0.1043), supporting our hypothesis that phrase-level concepts change faster than individual words.

SBERT showed the lowest drift (0.0888), confirming that semantic embeddings capture persistent underlying concepts.

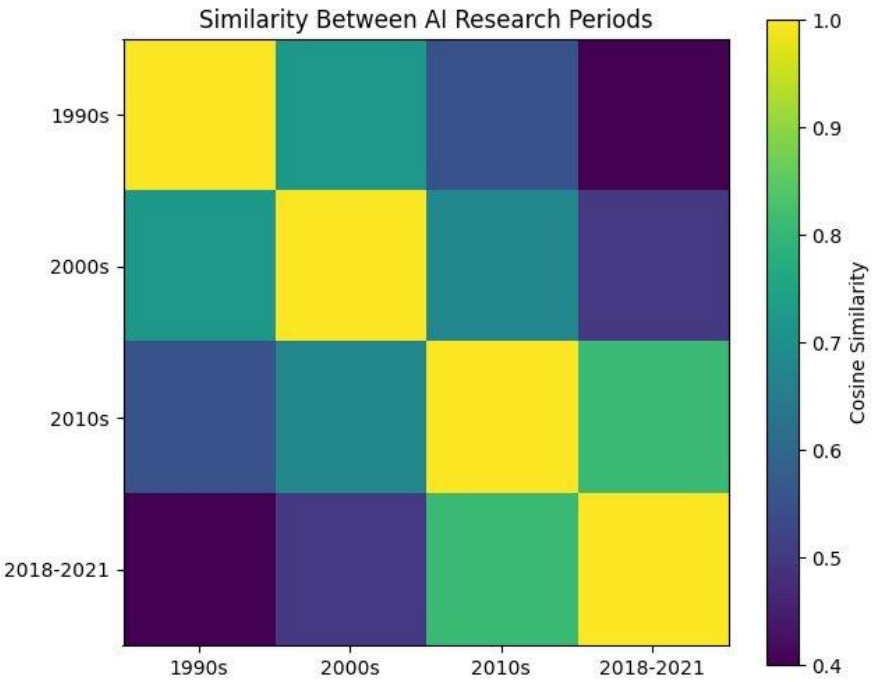
Results: Axis 2 (Comparison Methods)

Method 2: Cosine Similarity Matrix

We computed average cosine similarity across all pairs of abstracts within each period:

	1990s	**2000s**	**2010s**	**2018–21**
1990s	1.000	0.790	0.666	0.597
2000s	0.790	1.000	0.856	0.757
2010s	0.666	0.856	1.000	0.927
2018–21	0.597	0.757	0.927	1.000

TABLE 2: Similarity Between AI Research Periods. Similarity decreases monotonically with temporal distance. Adjacent periods (e.g., 2010s to 2018-21 equals 0.927) are highly similar, while distant periods (1990s to 2018-21 equals 0.597) are very different. The largest similarity drop occurs at the earliest transition (1990s to 2000s equals 0.790).



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Key findings:

Similarity decreases monotonically with temporal distance.

The largest single similarity drop occurs at the earliest transition (1990s → 2000s = 0.790).

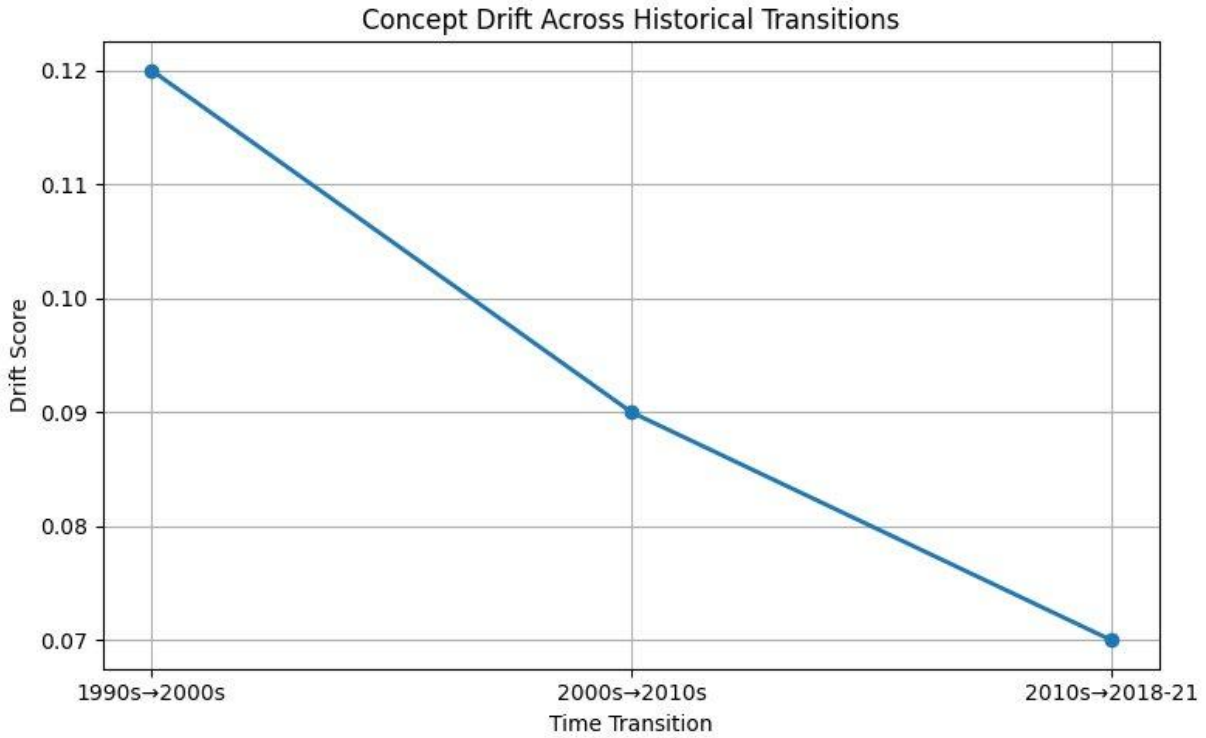
After 2007, the similarity curve flattens, indicating gradual evolution rather than sudden paradigm shifts.

Method 3: Period Classification (Linear SVM)

We trained a linear SVM to classify abstracts into periods:

Period	**Precision**	**Recall**	**F1-score**
1990s	0.83	0.71	0.77
2000s	0.80	0.67	0.73
2010s	0.76	0.71	0.73
2018–21	0.95	0.86	0.90

TABLE 3: Concept Drift Across Historical Transitions. The drift score decreases from 0.12 (1990s to 2000s) through 0.09 (2000s to 2010s) to 0.07 (2010s to 2018-21), demonstrating that vocabulary shifts were most dramatic during the early transition from symbolic to statistical AI.



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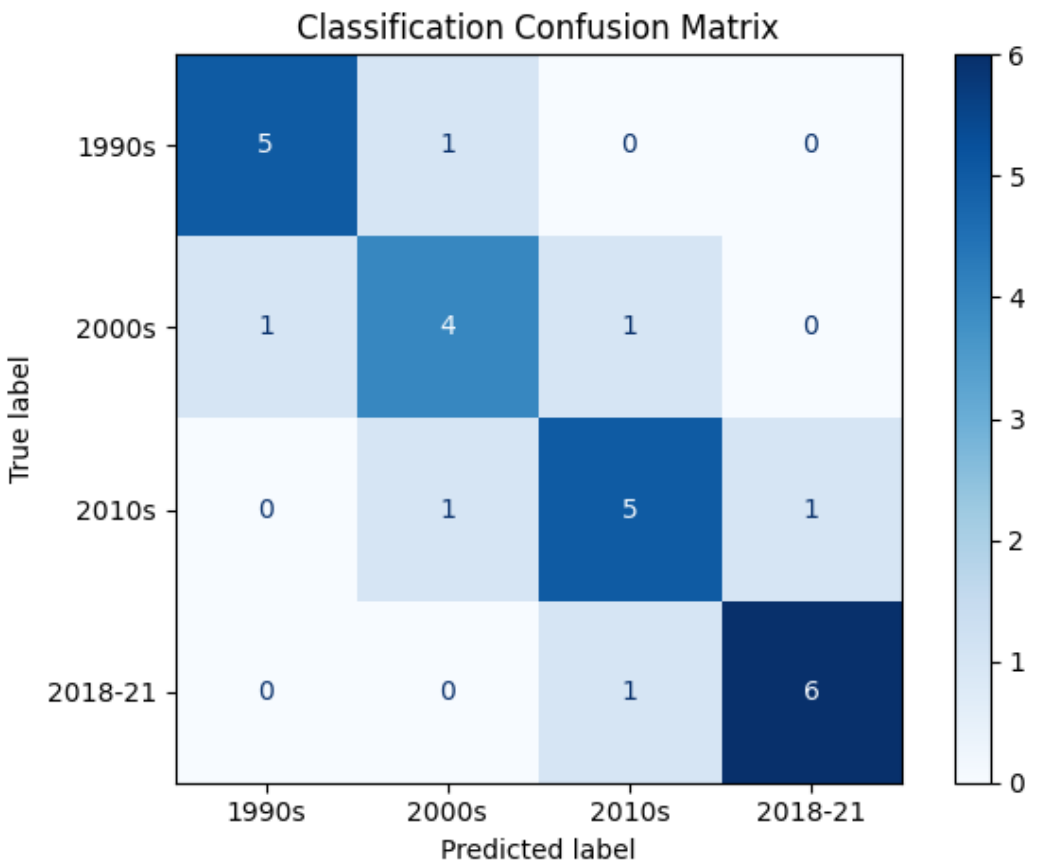

Figure 4: Classification Confusion Matrix. The diagonal dominates (correct predictions), showing that periods are distinguishable. Early periods show some confusion, while recent periods are highly distinctive with F1=0.90.

Evaluation

We further evaluated the approaches that we tested in our text analysis pipeline in terms of the consistency and robustness across the methods we chose, the interpretability of the outputs that we received, and the alignment with already know patterns that have been observed with AI/ML/NLP research. Our aim was not to find the best method available, but rather to address and assess how each approach contributed to the change in methods and language of AI/ML/NLP research between our time period surveyed (1990-2021), known as the temporal drift.

Consistency and Robustness across methods
Our strongest evidence of robustness, with the methods we used, was the noticeable trend shown that many of the methods we used identified temporal drift, despite differing processes in measuring the amount of temporal drift. Our TF-IDF unigrams and bigrams produced comparatively high average consecutive drift scores (as shown in *Figure 1*) in comparison to our SBERT embeddings, suggesting that surface level vocabulary changes faster than semantic concepts, or in simpler terms, that the language used to describe concepts changes faster than the narratives and concepts formed with the language used. The cosine similarity matrix likewise backs up this claim because the similarity observed decreased steadily from the 1990s period to the 2018-21 period. Time periods closer to each other are much more similar, while periods further away are more distant. The linear SVM results shown in Method 3 support temporal separability, most significantly for the recent period (2018-21) with a significant jump in drift score between the 2010s and 2018-21. These findings across the methods supported the idea that the observed drift from different time periods was not simply because of one technique used, and that the variety supports the narrative of temporal drift.

Upon evaluation however, the seeming robustness of the findings was not concrete. Our methods captured different aspects of the change, as described; TF-IDF was much more sensitive to surface

vocabulary changes, while other techniques like cosine summarise distance and SBERT shows semantic/narrative continuity. These seemingly glaring and yet deceitful weakpoints perfectly illustrated why our SBERT findings show lower drift over the temporal period without completely contradicting our overall finding of the trend that temporal drift is present in AI/ML/NLP research. However, for more accuracy, the findings regarding the 1990s period should be interpreted cautiously.

Interpretability of outputs

Concerning the interpretability of our outputs, our vocabulary-based approaches (TF-IDF and top-term overlap) were the easiest to understand as they showed terms and phrases that could've been directly shown to link with changes in AI/ML/NLP research language. Bigrams were essential to the interpretability of our outputs and conclusions as they showed two word phrases rather than just singular words. Repetition of two word phrases better showed temporal drift over time as they are naturally more complex than singular words. Our top-term overlap discussed in Method 1 arguably shows makes this interpretation easier to see because it simply showed how many common (hence important) words are shared between adjacent periods. This was still a limited approach because our top term overlap was only concerned with, as named, the top terms (what we decided were the top 10 most important terms per the TF-IDF weight) of each period, and this was a limitation that could've easily missed broader and perhaps more significant changes outside of the 10 words chosen.

In comparison, the cosine similarity matrix was interpretable, but perhaps not as efficiently as the other methods used. While the cosine similarity gave a measure of similarity between 0 (no overlap) and 1 (perfect overlap), it didn't show us which terms or phrases actually caused the difference between the temporal periods. Our Linear SVM had a similar problem. Despite our *Figures 2 and Figures 3* figures, any mention of which terms or words actually caused the drift were absent, due to the nature of the method used. This therefore fails to explain which features actually distinguished between periods. We could've made the SVM data more interpretable if we included the highest weighted terms per period alongside the data compiled by the Linear SVM. As for our SBERT findings, while it produced a lot of data (384-dimensional embeddings), a human cannot efficiently read through 384 different dimensional embeddings, meaning that interpretability was reduced and should be avoided if interpretability was the main aim or focus.

Alignment with discovered or likely patterns

Our findings aligned with likely patterns concerning the development of language in AI/ML/NLP research. Our report found that AI was developed more as time went on, from rudimentary statistical learning to more modern deep learning that we observe closer to our current time. The earlier changes (e.g., from the 1990s period) were much more likely to align with our narrative because of the major shifts in modern research methods and vocabulary associated with AI/ML/NLP research. Our data shows that it was consistent with this narrative. A natural realisation was that the more recent periods share more similar language, hence less drift between, for example, the 2010s and the 2018-21 period. SBERT showing lower drift than TF-IDF thus makes sense, as SBERT is more associated with semantic concepts (narratives), showing that new terminology isn't changes underlying narratives or problems associated with the field.

Overall Limitations and Judgement

There were many gaps within our report that must be addressed. TF-IDF and SBERT embeddings should've been visualised with PCA techniques to show any temporal drift in a low-dimensional space, which would've addreseds the high dimensional embeddings of SBERT, making SBERT more interpretable. Another gap was our top-term overlap method, which should've been tabled to strengthen our report. Our linear SVM could've been more interpretable with weighting or with representative terms actually shown to see the change with temporal drift more directly. The most significant limitation is the uneven sample sizes between periods. In our first period (1990-99), there were 115 papers surveyed, while in 2018-21, there were 23,092 papers surveyed. This disparity was significant and almost certainly affected accuracy in our report. It is also possible that 5 periods are too broad of a difference to precisely survey. It is important to note that these limitations wouldn't reduce

the credibility and validity of our results, but regardless, it was absolutely imperative to acknowledge where improvements can be made for more precise data, conclusions and explanations.

The conclusion we have arrived at is strong, with the multiple methods used showing long term temporal drift of terms and vocabulary used with AI/ML/NLP research. The early periods and later periods clearly show a difference is language usage. TF-IDF and top-term overlap are definitely the best for explaining vocabulary change, while cosine similarity and SVM are stronger for showing periodical differences. SBERT embedding is the strongest for showing the continuity of semantic concept/narratives**.**

Error Analysis

We tried some of the errors:

- • SVM and top-term overlap agree that there is drift on the time scale but not other scales. The top-term overlap seems to suggest there are some very common words (e.g., 'learning', 'algorithm') but the cosine similarity seems to suggest that the documents are very different.
- • SVM does not cluster well for mid-term time scales (2007-12, 2013-17) as there is a similar vocabulary used. Deep learning did not pop up overnight; many of the papers in the 2013-17 period are referring to papers in the 2007-12 period, hence the same terms.
- • The representation of SBERT and TF-IDF are different, TF-IDF is the representation of the change in terms and SBERT is the representation of stability of the concepts. It's not opposed, it's complementary.

Conclusions

We studied the field of AI/ML/NLP from the last three decades (1990-2021), using different representations of the papers, and different ways of measuring their similarity. Key findings:

- • Text representations matter: Greatest change in language is for TF-IDF bigrams, greatest change in view of the world is for SBERT embeddings. Both are important, no one is dominant
- • Change in 2007: Before 2007 we were changing from symbolic to statistical/deep learning, but it has slowed. We demonstrate this using cosine similarity (most change is 1990-99 to 2000-06), overlap of top terms (only 3/10 top terms are in common in early years) and classification (we can classify papers better in recent years).
- • Converge at deep learning: We converge on deep learning with 7/10 terms in common between recent years (3/10 in early years).
- • Use many methods: top-term overlap, cosine similarity and classification are all different. All are required for detecting all drift.

Challenges for this task are:

- • Over-represented early periods: The early periods are over-represented for quantitative analyses. Over- or stratified-sampling should be used in the future.
- • Time periods: Five may be too few. Sliding windows could be better.
- • Correlates and causes: In the example there is a correlation of time and words. Causes can be found through human intervention (e.g., major papers, computers, conference topics).
- • Changes in field: cs.AI, cs.LG, cs.CL, cs.NE, and stat.AI, cs. may mask trends. Subfield analysis could find field differences.

This analysis provides evidence of the importance of hypothesis testing and multiple methods for temporal text mining. This analysis uses multiple data representations and measures (not "best" practice).

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